Análisis Estadístico de Datos
Examen Argumentativo

Nombre: ______________________________ Matrícula:____________________

Instrucciones: Muestra cómo obtienes tu respuesta. Puedes usar Excel o Minitab para tus cálculos de probabilidades o percentiles, pero no uses otros archivos o hagas uso de la IA. Respeta el código de ética del curso y del Tecnológico de Monterrey.

1. A quality-control inspector examines 10 products. Each product has a probability of 0.30 of being defective, independently of the others.

   Let X denote the number of defective products.

   What is the probability that **exactly 4 products are defective**?

   (a) 0.1503
   (b) 0.8497
   (c) 0.2001
   (d) 0.6496

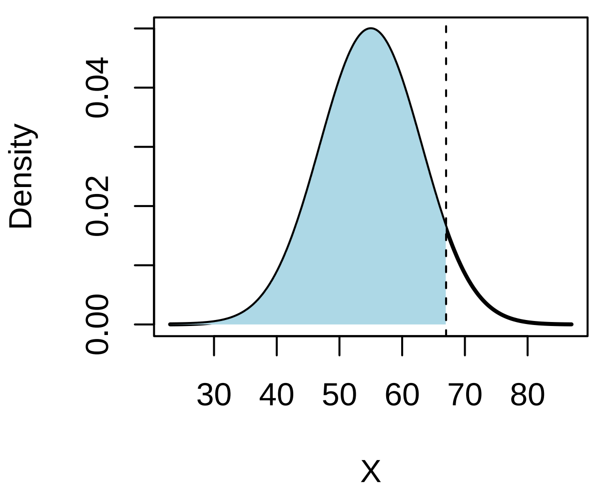
2. A random sample of size $n = 30$ was obtained from a population. The sample mean was $\bar{x} = 65$ and the sample standard deviation was $s = 10$.

   Compute the standard error of the sample mean using

$$SE(\bar{x}) = \frac{s}{\sqrt{n}}.$$

   Round your answer to two decimal places.

3. The following figure shows a normal distribution with mean $\mu = 55$ and standard deviation $\sigma = 8$. The shaded region represents $P(X \leq 67)$ for a normally distributed random variable X.



Compute the probability represented by the shaded area. Round your answer to four decimal places.